ASUS UGEN AI LEAGUE · STAGE I PROPOSAL

ShotSift

The cull happens *while you shoot*.

Creator AI | Battlefield Lightning · ASUS UGen300 | Onsite Track

Bryan Seah · Singapore Management University

github.com/hongyime/shotsift

Midnight after every event

Real numbers, from my own archive (02 CLIENT):

20
events shot
2023 – 2026

5,674
frames in the archive
(a subset of my shoots)

3–4 hrs
spent culling
per event

Every event ends the same way: **thousands of frames, one laptop, an evening gone to culling** — while the client's social team waits for photos that arrive the next day.

Three gaps no tool closes

⏱️

Timing

Every culling tool works **after** the shoot, on the photographer's laptop. Nothing helps **during** the event, when the client's comms team needs content.

📷📷

Multi-shooter

Several photographers cover one event; frames scatter across memory cards and get merged by hand. Nobody's photos are in one place until long after.

🔒

Privacy

Corporate clients don't want attendee faces uploaded to a cloud service — a real PDPA concern. Cloud culling is a contractual problem, not a convenience.

Who feels it



The event photographer

"I shot 2,000 frames tonight and I'll spend tomorrow morning culling instead of booking the next job."



The client's comms team

"The keynote ended two hours ago and we have nothing to post. Our moment is gone."



The AV / production house

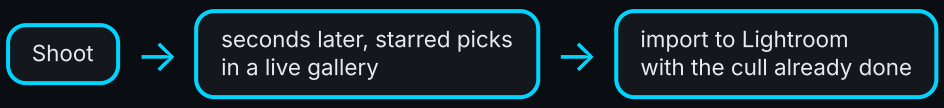
"I coordinate three shooters and the deliverables land as three separate piles I have to reconcile."

Existing tools — and the gap

	AfterShoot	Narrative Select	FilterPixel	ShotSift
When	After shoot	After shoot	After shoot	**During the shoot**
Where it runs	On-device	On-device	Cloud	**On-device (edge)**
Multi-shooter merge	No	No	No	**Yes**
Personal taste	AI profiles[1]	AI presets[1]	Generic	**On-device taste model[2]**

[1] Requires 1,500–5,000 previously edited images in a Lightroom catalog (per vendor sites). [2] Stage II roadmap — trained on-device on the photographer's own keep/reject decisions, no cloud. Competitor data from vendor websites, verified Oct 2026 (see appendix A3). No tool above processes frames during the shoot.

ShotSift in one sentence



An edge appliance — host + ASUS UGen300 — that scores every frame as it lands and writes the cull into XMP sidecars your editor already understands.

User journey



Before the event

Plug in the box and a venue-local access point. Cameras connect over FTP-over-Wi-Fi — the feature most pro bodies already have. Card reader as fallback.



During the event

Frames arrive as they are shot. The client's social team browses a live gallery on the local network — captioned picks, while the event is still running.



After the event

XMP sidecars carry star ratings, labels and keywords next to the RAWs. Import to Lightroom or Capture One — the cull is already there. Event data auto-deletes.

Innovation 1 — it learns your taste



Generic "aesthetic" scores don't match how professionals choose. A taste model trained on **your own decisions** culls the way **you** cull.

> Honest framing: taste-model training is **Stage II** work. Stage I demonstrates the full pipeline with heuristic weights (focus + exposure). No rating history was available for Stage I — and none was fabricated.

Innovation 2 — three-tier cascade, one chip

TIER 1 · every frame · Hailo-10H vision

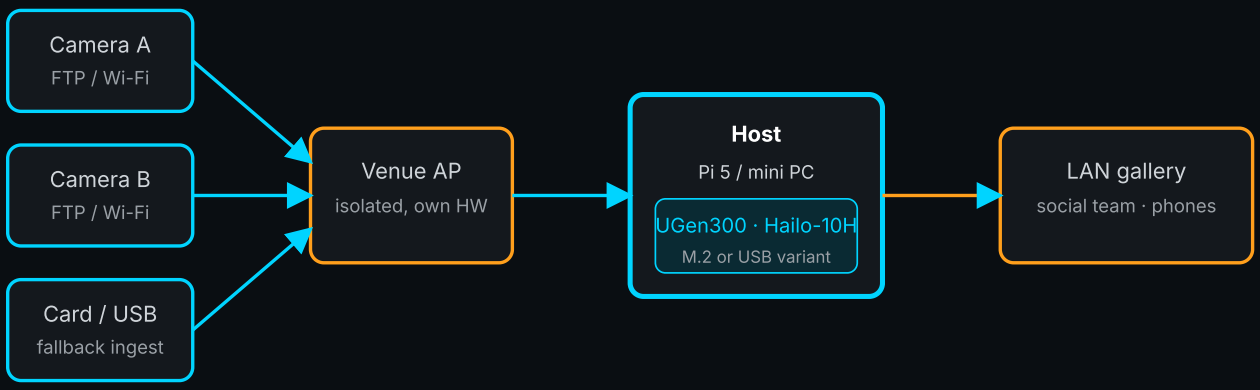
SCRFD face detection → ArcFace embeddings (identity clustering) · 68-pt landmarks → blink check · CLIP image encoder → semantic + aesthetic

TIER 2 · top picks only · Hailo-10H GenAI

Qwen2-VL-2B → captions + keywords · Qwen2-1.5B function-calling → natural-language search ("speaker laughing with the CEO")

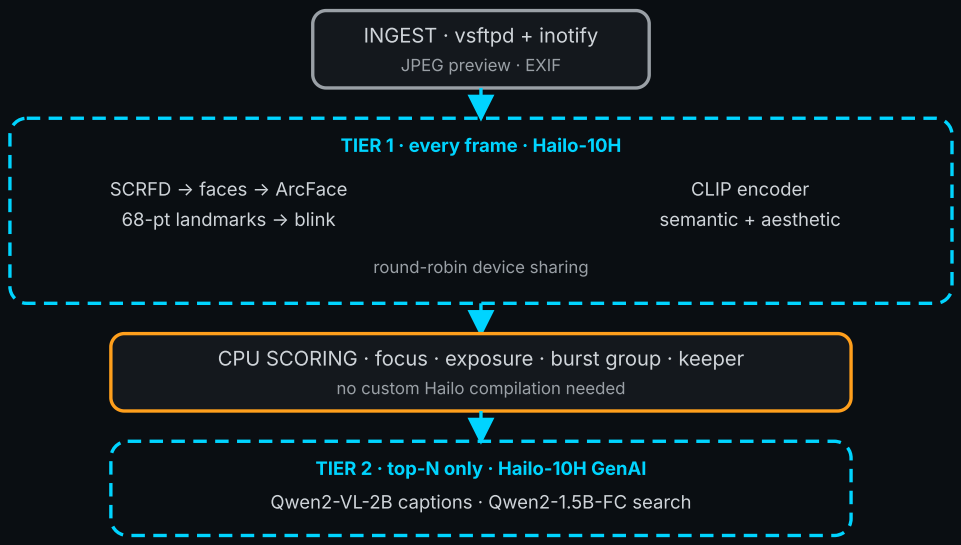
One UGen300, shared by round-robin scheduling. Custom logic stays on the CPU — **no Hailo model compilation required**.

Hardware architecture



All venue-local. Bring your own access point — never depend on crowded house Wi-Fi. Nothing leaves the venue.

Software architecture



Outputs: XMP sidecars · FastAPI LAN gallery · SQLite · Docker Compose · event data wiped after delivery

Model map

Model	Task	Runs on	Source
SCRFD	Face detection	Hailo-10H	Hailo Model Zoo
ArcFace	Identity embeddings	Hailo-10H	Hailo Model Zoo
3DDFA	68-pt landmarks → blink	Hailo-10H	Hailo Model Zoo
CLIP (image)	Semantic + aesthetic	Hailo-10H	Hailo Model Zoo
Qwen2-VL-2B	Captions + keywords	Hailo-10H	Hailo GenAI Zoo
Qwen2-1.5B-FC	Natural-language search	Hailo-10H	Hailo GenAI Zoo
Heuristics + taste head	Focus / exposure / burst / keeper	CPU	Custom (this repo)

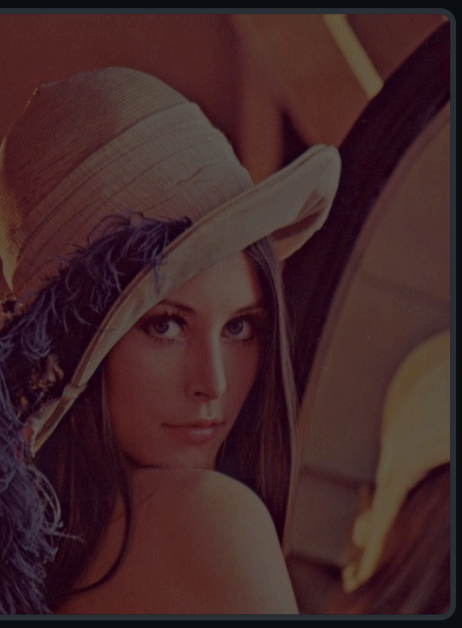
Only stock precompiled models run on the Hailo. Custom logic stays on CPU — **no custom compilation required**. Strong feasibility signal for a 4-week build.

Privacy by design

- **No cloud.** Every byte is processed on-device or on the venue LAN.
- **Scoped embeddings.** Face embeddings live for one event, then auto-delete.
- **RAWs never leave the venue.** Only embedded JPEG previews are processed.
- **Opt-in search.** The "find my photos" selfie kiosk has a consent screen.

Biometric data is personal data under Singapore's PDPA — ShotSift is architected so there is nothing to breach.

Stage I prototype — what works today



underexposed → rejected



soft focus → flagged



★ sharp → 4-star keeper

Demonstrated: burst grouping · focus scoring · keeper ranking · star ratings · XMP sidecars · HTML gallery — verified end-to-end on a 3-frame burst.

Stage II: face detection · blink detection · CLIP · captions · search · taste model.

Expected outcomes — validation plan

Targets for Stage II validation, not claims:

Metric	Target / method
Shutter-to-gallery latency	Measured capture-timestamp → gallery render
Sustained throughput	FPS under multi-shooter burst load
Taste-model precision@K	vs. historical picks, held out by whole event
Duplicate collapse rate	Share of frames folded into burst groups
Power	Watts at the wall
Time saved	Manual cull vs. ShotSift-assisted, same event

Four-week Stage II plan

Week 1

Port Tier 1 to the Hailo-10H. Benchmark per-model latency; confirm zoo model versions against HailoRT.

Week 2

GenAI tier (Qwen2-VL-2B, 1.5B-FC) + round-robin device sharing across all models.

Week 3

Live gallery, XMP writer, multi-camera FTP ingest. Docker Compose packaging.

Week 4

Validation runs on two real shoots. Documentation, demo rehearsal, submission.

Risks and mitigations

Risk	Mitigation
Model missing from Hailo-10H zoo	CPU fallback via ONNX Runtime; noted in deck
VLM latency (seconds/image)	Background queue; captions for top picks only
Venue Wi-Fi unreliability	Own access point; Ethernet for the host
Taste-model overfitting	Event-level holdout + generic baseline comparison
Hardware arrival timing	Host-agnostic Docker build; prototype runs on any laptop

Business model



1 · AV rental add-on

Per-event rental through AV and event-production houses — a line item on the kit list. No consumer sales cycle; I know this world.



2 · Studio subscription

Per-seat subscription for photography studios running multi-shooter events — taste model + gallery hosting per seat.



3 · ASUS channel

Reference software for a ProArt laptop + UGen300 creator bundle. A product ASUS's own business could sell.

Go-to-market — and why me

- **First pilots** from my own photographer and AV network — youth, community and corporate events I already shoot and crew.
- **Zero workflow change:** XMP sidecars mean Lightroom / Capture One users adopt it without learning anything new.
- **Why me:** a working event photographer and AV operator who also builds self-hosted AI infrastructure — I live both sides of this problem.

References and links

▸ github.com/hongyime/shotsift — prototype + docs

▸ Demo video (unlisted YouTube link in submission)

▸ Hailo Model Zoo — hailo.ai/developer-zone (SCRFD, ArcFace, CLIP)

▸ Hailo GenAI Zoo (Qwen2-VL-2B, Qwen2-1.5B-FC)

▸ Deng et al., "ArcFace" (CVPR 2019); Zhang et al., SCRFD

▸ Radford et al., "CLIP" (ICML 2021)

▸ Qwen2-VL technical report (2024)

▸ McInnes et al., HDBSCAN (2017)

▸ Adobe XMP Specification

▸ PDPC Singapore — guidance on biometric data

▸ aftershoot.com · narrative.so · filterpixel.com (competitor data, Oct 2026)

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